

Kessence'particles'ID'unequal'mass'mu

by Miguel Ramiro

Version: 23

January 28, 2021

Fields

Psisf, phi, Phid_x, Phid_y, Phid_z, Psi

Spatial Coordinates

x, y, z

Time Coordinate

t

Auxiliary Fields

phi2, dphi2

Auxiliary Variables

r1_x, r1_y, r1_z, r2_x, r2_y, r2_z, Xphi, Kf, KXf, KXXf, Phiu_x, Phiu_y, Phiu_z, Tbar, sq_r1, sq_r2, rorbit_dynamic, omegaorbit, nu_spatial, radius, gammax_off

Parameters

Parameter	Type	Default value
<i>Mpl_cte</i>	REAL	Not set
<i>sigmax</i>	REAL	Not set
<i>betax</i>	REAL	Not set
<i>gammax</i>	REAL	Not set
<i>p_sigma</i>	REAL	Not set
<i>max_error</i>	REAL	Not set
<i>p_Omega</i>	REAL	Not set
<i>Aphi1</i>	REAL	Not set
<i>Aphio</i>	REAL	Not set
<i>tini</i>	REAL	0.0
<i>tslow</i>	REAL	Not set
<i>torbit</i>	REAL	800.0
<i>tslow2</i>	REAL	2.0
<i>sigmaslow</i>	REAL	10.0
<i>rorbit</i>	REAL	Not set
<i>rstart</i>	REAL	150.0
<i>mass</i>	REAL	Not set
<i>massfactor</i>	REAL	Not set
<i>xcenter1</i>	REAL	0
<i>ycenter1</i>	REAL	0
<i>zcenter1</i>	REAL	0
<i>rdonut</i>	REAL	4.0
<i>Nstar</i>	INT	1
<i>xf</i>	REAL	15
<i>wd</i>	REAL	0.5
<i>nu</i>	REAL	1.0
<i>nu_asymp</i>	REAL	0.01
<i>nu_radius</i>	REAL	1500
<i>nu_width</i>	REAL	0.02
<i>offw</i>	REAL	1.0
<i>timehyper</i>	REAL	2000
<i>mu</i>	REAL	0.5

Evolution Equations

Evolution equation

$$\partial_t Psisf = S^{(Psisf)} + Op(x, y, z, t) \quad (1)$$

$$\begin{aligned} Op(x, y, z, t) = & + \frac{2}{KXf} KXXf Phid_z^2 \partial_z Phid_z \\ & + \frac{4}{KXf} KXXf Phid_z Phid_y \partial_z Phid_y \\ & + \frac{4}{KXf} KXXf Phid_z Phid_x \partial_z Phid_x \\ & + \frac{2}{KXf} KXXf Phid_y^2 \partial_y Phid_y \\ & + \frac{4}{KXf} KXXf Phid_y Phid_x \partial_y Phid_x \\ & + \frac{2}{KXf} KXXf Phid_x^2 \partial_x Phid_x + \partial_z Phid_z \\ & + \partial_y Phid_y + \partial_x Phid_x + \left(-\frac{1}{Mpl_cte} \right) \frac{\frac{Tbar}{KXf}}{4} \end{aligned} \quad (2)$$

Evolution equation

$$\partial_t Psisf = S^{(Psisf)} + Op(x, y, z, t) \quad (3)$$

$$Op(x, y, z, t) = +0.0 \quad (4)$$

Evolution equation

$$\partial_t phi = S^{(phi)} + Op(x, y, z, t) \quad (5)$$

$$Op(x, y, z, t) = +Psi \quad (6)$$

Evolution equation

$$\partial_t Phid_x = S^{(Phid_x)} + Op(x, y, z, t) \quad (7)$$

$$Op(x, y, z, t) = +\partial_x Psisf - Phid_x \nu \quad (8)$$

Evolution equation

$$\partial_t Phid_x = S^{(Phid_x)} + Op(x, y, z, t) \quad (9)$$

$$Op(x, y, z, t) = +\partial_x Psi \quad (10)$$

Evolution equation

$$\partial_t Phid_y = S^{(Phid_y)} + Op(x, y, z, t) \quad (11)$$

$$Op(x, y, z, t) = +\partial_y Psisf - Phid_y \nu \quad (12)$$

Evolution equation

$$\partial_t Phid_y = S^{(Phid_y)} + Op(x, y, z, t) \quad (13)$$

$$Op(x, y, z, t) = +\partial_y Psi \quad (14)$$

Evolution equation

$$\partial_t Phid_z = S^{(Phid_z)} + Op(x, y, z, t) \quad (15)$$

$$Op(x, y, z, t) = +\partial_z Psisf - Phid_z \nu \quad (16)$$

Evolution equation

$$\partial_t Phid_z = S^{(Phid_z)} + Op(x, y, z, t) \quad (17)$$

$$Op(x, y, z, t) = +\partial_z Psi \quad (18)$$

Evolution equation

$$\partial_t Psi = S^{(Psi)} + Op(x, y, z, t) \quad (19)$$

$$\begin{aligned} Op(x, y, z, t) = & -\frac{2}{2 Psi^2 - \frac{KXf}{KXXf}} Phid_x^2 \partial_x Phid_x \\ & -\frac{4}{2 Psi^2 - \frac{KXf}{KXXf}} Phid_y Phid_x \partial_y Phid_x \\ & -\frac{4}{2 Psi^2 - \frac{KXf}{KXXf}} Phid_z Phid_x \partial_z Phid_x \\ & +\frac{4}{2 Psi^2 - \frac{KXf}{KXXf}} Phid_x Psi \partial_x Psi \\ & -\frac{2}{2 Psi^2 - \frac{KXf}{KXXf}} Phid_y^2 \partial_y Phid_y \\ & -\frac{4}{2 Psi^2 - \frac{KXf}{KXXf}} Phid_z Phid_y \partial_z Phid_y \\ & +\frac{4}{2 Psi^2 - \frac{KXf}{KXXf}} Phid_y Psi \partial_y Psi \\ & -\frac{2}{2 Psi^2 - \frac{KXf}{KXXf}} Phid_z^2 \partial_z Phid_z \\ & +\frac{4}{2 Psi^2 - \frac{KXf}{KXXf}} Phid_z Psi \partial_z Psi \\ & -\frac{1}{2 \frac{KXXf}{KXf} Psi^2 - 1.0} \partial_z Phid_z - \frac{1}{2 \frac{KXXf}{KXf} Psi^2 - 1.0} \partial_y Phid_y \\ & -\frac{1}{2 \frac{KXXf}{KXf} Psi^2 - 1.0} \partial_x Phid_x + \frac{\frac{1}{Mpl_cte}}{2 KXXf Psi^2 - KXf} \frac{Tbar}{4} \end{aligned} \quad (20)$$

Evolution equation

$$\partial_t Psi = S^{(Psi)} + Op(x, y, z, t) \quad (21)$$

$$Op(x, y, z, t) = +0.0 \quad (22)$$

Auxiliary Field Equations

$$phi2 = phi \ phi \tag{23}$$

$$dphi2 = Phid_x \ Phid_x + Phid_y \ Phid_y + Phid_z \ Phid_z \tag{24}$$

Auxiliary Variable Equations

Auxiliary variable equation

$$r1_x = \tag{25}$$

Auxiliary variable equation

$$r1_y = \tag{26}$$

Auxiliary variable equation

$$r1_z = \tag{27}$$

Auxiliary variable equation

$$sq_r1 = \tag{28}$$

Auxiliary variable equation

$$r2_x = \tag{29}$$

Auxiliary variable equation

$$r2_y = \tag{30}$$

Auxiliary variable equation

$$r2_z = \tag{31}$$

Auxiliary variable equation

$$sq_r2 = \tag{32}$$

Auxiliary variable equation

$$Tbar = \tag{33}$$

Auxiliary variable equation

$$Xphi = \tag{34}$$

Auxiliary variable equation

$$Kf = \tag{35}$$

Auxiliary variable equation

$$KXf = \tag{36}$$

Auxiliary variable equation

$$KXXf = \tag{37}$$

Auxiliary variable equation

$$Phiu_x = \tag{38}$$

Auxiliary variable equation

$$Phiu_y = \tag{39}$$

Auxiliary variable equation

$$\textit{Phiu}_z = \tag{40}$$

Auxiliary variable equation

$$\textit{omegaorbit} = \tag{41}$$

Auxiliary variable equation

$$\textit{rorbit_dynamic} = \tag{42}$$

Auxiliary variable equation

$$\textit{nu_spatial} = \tag{43}$$

Auxiliary variable equation

$$\textit{radius} = \tag{44}$$

Auxiliary variable equation

$$\textit{gamma}_{\textit{off}} = \tag{45}$$